

Phishing Detection & Awareness Report

This report demonstrates phishing email analysis, threat identification, email header investigation, risk classification, and employee awareness practices used to reduce phishing-related cyber threats in organizations.

Project Objectives

Analyze phishing email samples
Identify phishing indicators
Investigate suspicious domains and URLs
Classify email threats
Create awareness guidelines for employees

Tools Used

Tool	Purpose
Google Header Analyzer	Email header investigation
MXToolbox	SPF/DKIM analysis
VirusTotal	Malicious URL scanning
GitHub	Project hosting

Common Phishing Indicators

Fake or suspicious sender domains
Urgency and fear-based language
Malicious links or fake login pages
Generic greetings such as “Dear User”
Dangerous attachments

Email Risk Classification

Classification	Description
Safe	Legitimate sender and trusted content
Suspicious	Contains warning signs
Phishing	Malicious intent detected

Phishing Awareness Infographics

phishing-email.png

Subject: ⚠️ Urgent: Your Account Will Be Locked

Security Team <security@secure-account-verify.com> **Suspicious Sender Domain**

To: You

Today, 10:24 AM

Dear User, **Generic Greeting**

We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

Verify Now: <http://secure-account-verify.com> **Suspicious Link (Do not click)**

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

⚠️ This email is a phishing attempt. Do not click links or share your information.

fake-domain-analysis.png

FAKE DOMAIN ANALYSIS

LEGITIMATE DOMAIN	FAKE DOMAIN
 paypal.com https://www.paypal.com	 paypa1.com http://secure-paypal-login.com
✓ Uses correct spelling ✓ Uses HTTPS (Secure) ✓ Official and trusted domain ✓ Used by the official company	✗ Uses number "1" instead of letter "l" ✗ Does not use HTTPS (Not secure) ✗ Untrusted and fake domain ✗ Created to steal your information
 SAFE	 DANGEROUS

💡 Always check the domain carefully. Attackers use fake domains that look similar to real ones.

header-analysis.png

EMAIL HEADER ANALYSIS

Header Field	Value	Status	Explanation
Return-Path	<security@secure-account-verify.com>	✗ FAIL	Does not match the official domain
SPF	FAIL (secure-account-verify.com)	✗ FAIL	Sender not authorized by domain
DKIM	FAIL	✗ FAIL	Email not signed by the domain owner
DMARC	FAIL	✗ FAIL	Domain not protected by DMARC policy
Received	from unknown (185.199.108.153)	⚠️ SUSPICIOUS	Email received from suspicious server
Message-ID	<202405141024.secure@randommail.com>	⚠️ SUSPICIOUS	Message ID from a random domain

✗ This email failed multiple authentication checks and appears to be a phishing email.

awareness-guidelines.png

PHISHING AWARENESS GUIDELINES

DO'S ✓	DON'TS ✗
 Verify the sender's email address before trusting the email.	 Don't open attachments from unknown senders.
 Hover over links to check the actual URL before clicking.	 Don't click on links in suspicious or unexpected emails.
 Report suspicious emails to your IT or security team.	 Don't share your passwords, OTPs, or personal data.
 Enable Multi-Factor Authentication (MFA) wherever possible.	 Don't respond to threatening or urgent emails.
 Keep your passwords and personal information confidential.	 Don't ignore or delete security warnings.

 **Stay Alert. Stay Safe. Think Before You Click!**

Prevention Guidelines

Verify sender email addresses carefully Hover over links before clicking Do not open suspicious attachments Enable Multi-Factor Authentication (MFA) Report suspicious emails immediately

Conclusion

This project successfully demonstrated phishing detection techniques through email analysis, header investigation, fake domain identification, and awareness reporting. The report highlights how phishing attacks rely on social engineering and why employee awareness is critical in cybersecurity.